

# Mobina Pournemat

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📍 Mountain View, California

## OVERVIEW

I am a third-year Computer Science Ph.D. student at the University of Maryland, College Park, advised by Prof. Soheil Feizi. My research centers on reasoning in large language models — in particular **efficient inference and reasoning**, **reasoning faithfulness**, and the **evaluation** frameworks. I also have experience in multimodal generative models, computer vision, and applied machine learning in industry.

## EDUCATION

- **University of Maryland, College Park** Maryland, USA  
*Ph.D. in Computer Science; Advised by Prof. Soheil Feizi; GPA: 4/4* Aug. 2024 - Present
- **Sharif University of Technology** Tehran, Iran  
*B.Sc. in Computer Engineering* Sep. 2018 - Feb. 2023

## RESEARCH EXPERIENCE

- **University of Maryland, College Park** Maryland, USA  
*Research Assistant; Advisor: Prof. Soheil Feizi* Aug. 2024 - Present
  - **Efficient Reasoning through Cross-Model Latent Knowledge Transfer (in progress):**  
Reducing long chain-of-thought inference cost by transferring reasoning knowledge across models in latent space.
  - **Exploring LLMs' Probabilistic Reasoning Capabilities (EMNLP 2026, In Collaboration with Meta AI):**  
Built a comprehensive benchmark for reasoning under uncertainty and uncovered systematic failure modes.
  - **Evaluating LLM Reliability Beyond Accuracy:**  
Quantified how consistently LLMs answer semantically identical questions.
  - **Fine-Grained Instruction Control for Image Editing (CVPR 2026):**  
Contributed to SliderEdit, a continuous, per-instruction control interface for instruction-based image editing.
- **Institute of Science and Technology Austria (ISTA)** Klosterneuburg, Austria  
*Machine Learning Intern; Supervisor: Prof. Johann Danzl* Nov. 2023 - Jul. 2024
  - **3D Semantic Segmentation, Microscopy Image Analysis:**  
Developed an efficient pipeline for automated semantic segmentation of brain microscopy images using *Segment Anything*.
- **Sharif University of Technology** Tehran, Iran  
*NLP Research Assistant; Advisor: Dr. Ehsaneddin Asgari* Jul. 2022 - Oct. 2023
  - **Persian Syntactic and Semantic Error Correction Tool:**  
Developed an auto-correction tool for Persian text, combining a BERT-based language model with rule-based modules.
- **Sharif University of Technology** Tehran, Iran  
*Computer Vision Research Assistant; Advisor: Dr. Mohammad Hossein Rohban* Sep. 2021 - Jul. 2022
  - **Anomaly Detection with Autoencoders; Semantic Segmentation:**  
Benchmarked reconstruction- and representation-based methods on standard vision datasets.

## PUBLICATIONS

- **Reasoning Under Uncertainty: Exploring Probabilistic Reasoning Capabilities of LLMs**  
*M Pournemat, K Rezaei, G Sriramanan, A Zarei, H Eghbalzadeh, Y Wang, J Fu, F Peng, Z Wang, X Feng, N Noorshams, S Feizi - (EMNLP 2026, Main Conference)*
- **SliderEdit: Continuous Image Editing with Fine-Grained Instruction Control**  
*A Zarei, S Basu, M Pournemat, S Nag, R Rossi, S Feizi - (CVPR 2026, Oral Presentation)*
- **Same Question, Different Answers: Evaluating LLM Reliability Beyond Accuracy**  
*K Faghieh, Y Cheng, S Saha, M Pournemat, A Gerami, S Feizi - (Under Review)*

## WORK EXPERIENCE

- **Behin Bazaar Marketing Agency** Tehran, Iran  
*Data Analyst, Business Intelligence* *Apr. 2023 - Oct. 2023*
  - **Data Analysis, Database Design, Data Preprocessing, Visualization, Reporting:**  
Designed the analytics database and delivered SQL and Power BI dashboards used for campaign decisions.
- **AIMedic** Tehran, Iran  
*Machine Learning for Healthcare Intern* *Jul. 2021 - Sept. 2021*
  - **Data Science, Classification, Semantic Segmentation:**  
Built and evaluated classification and segmentation models for medical imaging data.

## SELECTED PROJECTS

- **Quantifying Attributional Bias in LLM Toxicity Detection** (Nov. 2025):  
Measuring how toxicity scores for identical comments change when attributed to different author identities.
- **Benchmarking Multi-Stage Vulnerability Reasoning in LLMs** (Oct. 2025):  
Evaluating the effect of explicit reasoning and post-verification on LLM performance in vulnerability detection.
- **Explainable Detection of Online Sexism** (May 2022):  
SemEval-2023, Task 10: hierarchical classification and explanation of sexist content.
- **Recommender System Design** (May 2021):  
Content-based, collaborative filtering, and hybrid recommenders with offline evaluation of ranking quality.

## TEACHING ASSISTANT EXPERIENCE

- **Introduction to Artificial Intelligence:** Fall 2024
- **Data Structures and Algorithms:** Fall 2022
- **Natural Language Processing:** Spring 2022
- **Computer Simulation:** Fall 2021
- **Probability and Statistics:** Spring 2021

## RELEVANT COURSEWORK

- **Natural Language Processing:** 4/4 (A)
- **Long-Context Language Models:** 4/4 (A)
- **Multimodal Foundation Models:** 4/4 (A)
- **Large Language Models Security and Privacy:** 4/4 (A)
- **Advanced Numerical Optimization:** 4/4 (A)
- **Artificial Intelligence:** 20/20
- **Data Structures and Algorithms:** 19.6/20
- **Stanford CS229 - Machine Learning:** online, audited
- **Natural Language Processing Specialization:** Coursera

## TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, SQL, JavaScript, PHP
- **Machine Learning Libraries:** PyTorch, Hugging Face Transformers, TensorFlow Keras, NumPy, Pandas, scikit-learn
- **Web Development & Database:** HTML/CSS, jQuery, Bootstrap, MySQL, PostgreSQL, SQL Server
- **Miscellaneous:** Microsoft Office, Power BI, Docker, Git, Linux, L<sup>A</sup>T<sub>E</sub>X

## LANGUAGES

**Persian:** native     **English:** proficient     **Spanish:** low proficiency     **Turkish:** low proficiency